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Hybrid encryption scheme based on temporal ghost imaging

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Abstract
In this study, we propose an optical hybrid encryption scheme based on compressive temporal ghost imaging. This scheme combines the asymmetric of the public key cryptography and the non-linear of ghost imaging. First, according to the principle of temporal ghost image encryption, the chaotic modulation pattern is used as the key to encrypt the plaintext information, and then the chaotic pattern is encrypted by the public key cryptography. Then, only the authorized receiver can decrypt the chaotic pattern with the private key. Finally, the plaintext can be decrypted with the compressive sensing algorithm. This scheme solves the problem of key distribution in traditional ghost imaging encryption scheme. At the same time, the application of public key cryptography reduces the cost of establishing private channels and enhances the security of the system. In addition, feasibility, security, and robustness of the scheme were verified by simulation. This scheme has the advantages of high quality, high safety, wide applicability and low cost, which can immediately be applied to encryption.

1 Introduction

With the deepening of economic globalization and informatization, information exchanges become more and more frequent, following that the security of information has been necessary in real life. Optical encryption is attracting much attention due to its advantages of multi-dimension, high speed, and parallelism [1]. Optical encryption mainly includes two types. One is encrypting the image by optical transformation, such as the double random-phase encoding (DRPE) based on the optical 4f setup [2–5]. The other is using light as transmission medium for secure optical communication, such as chaotic optical communication and quantum communication [6–8]. Recent reports have pointed out that DRPE systems may be cracked [9, 10]. At the same time, although quantum communication has greatly improved information security, the system is complex. Therefore, a simple and easy-to-use optical encryption and transmission method is urgently needed.

Ghost imaging [GI] is a technique that allows for constructing the image of an object without actually seeing the object. Since 1995, Pittman achieved ghost imaging with entangled photon pairs, the GI has a rapid development [11–15]. In 2010, Pere Clemente proposed optical encryption based on the GI [16]. In this scheme, a random light field is illuminated on a secret image repeatedly, and then the transmitted light is recorded by a bucket detector to obtain a ciphertext. The secret image is retrieved by correlating the ciphertext with random light fields. In the past few years, optical encryption-based GI has achieved rapid development. This approach has the capability of encrypting ghost images by flexibly manipulating the position correlation of a pair of "signal" and "idler" beams [17]. In addition, Chen W. proposed a series of optical encryption methods based on GI [18–20]. In paper [21], higher security and better robustness optical encryption based on GI and QR code have been proposed.

For multiple-image encryption, Wu et al. proposed an optical multiple-image encryption scheme based on GI with the position multiplexing [22]. Meng et al. came up with multiple-image encryption based on GI, (t, n) threshold secret sharing, and phase retrieval in the Fresnel domain [23]. By exploiting the duality between light propagation in space and time, Piotr Ryczkowski brought up temporal ghost imaging (TGI), and achieved temporal resolution at the picosecond level [24, 25]. The temporal ghost imaging provided a new idea for ultra-fast signal detection and optical communication. Pan et al. used compressive temporal ghost imaging with a chaotic laser to encrypt temporal signals [26]. Liu

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et al. pointed out that temporal ghost imaging can detect the temporal signal using a receiver far below the signal bandwidth [27]. This provides a powerful basis for temporal ghost imaging to achieve high-speed transmission. However, optical encryption methods based on GI or TGI have some problems, such as, if the receiver wants to decrypt the ciphertext, it must receive a large number of random light field patterns. At the same time to ensure the key security, whether it is to establish a private channel or take additional security measures will increase the cost. These problems affect the security and efficiency of encryption methods.

In this work, we propose an optical hybrid encryption scheme based on compressive temporal ghost imaging (HTGI). In the system, the encipherer uses temporal ghost imaging with a chaotic laser to encrypt the signal, in which the chaotic laser parameters are encrypted by the public key of the RSA public key algorithm. The encipherer transmits ciphertexts to the receiver through the common channel. The receiver uses the private key to decrypt the encrypted chaotic laser parameters and then uses compressive sensing to decrypt the encrypted signal. Simulation results verify the feasibility, security, and robustness of the proposed encryption scheme. This scheme uses temporal ghost imaging to encrypt the plaintext and uses public key cryptography to encrypt the key. This scheme as a hybrid encryption method improves the security of information, solves the problem of key distribution, and reduces the additional cost of establishing private channels. Even if the ciphertext is stolen, it is difficult to get useful information. However, it is worth noting that this scheme has limitations when coping with quantum computing or quantum eavesdroppers. In a word, this scheme provides a new way for encryption and telecommunication with the advantages of high security, strong robustness, and high quality of reconstructed information.

2 Theory analysis

2.1 Optical encryption based on compressive temporal ghost imaging

Compared to traditional ghost imaging, temporal ghost imaging uses a non-coherent light source of random temporal flotations to detect the temporal objects. Compressive sensing can significantly reduce the measurement time and improve the reconstructed signal quality. Recently, many researchers used compressive ghost imaging encryption to overcome the security problems of linear encryption systems. However, if we want to decrypt the information, regardless of ghost imaging or temporal ghost imaging, precise information of light field intensity distribution is required. It leads to a huge, redundant key space in cryptographic system, the efficiency of transmission is reduced greatly. In this work, we propose improved optical encryption based on compressive temporal ghost imaging with a chaotic laser.

The chaotic laser has good time-controllable random characteristics and a second-order time correlation property can be used as the light of the temporal ghost imaging. The generation of chaotic light is mainly dependent on the setting of the chaotic model and initial parameters. The chaotic light is difficult to reproduce when the model and initial parameters are not known. In this study, we use a simple but widely used Logistic model. Logistic mapping is defined as follows:

$$x_{n+1} = \mu x_n(1 - x_n), \tag{1}$$

where x_n is iterative initial value, $x_n \in [0, 1]$. μ is the system parameter, $\mu \in [0, 4]$.

The optical cryptography based on temporal ghost imaging with a chaotic laser is shown in Fig. 1. Let us suppose that Alice wants to transmit encrypted information of an object to Bob. First, Alice uses chaotic parameters which are produced by the pseudo-random number generator to generate the chaotic laser. The chaotic laser $I(t)$ through the temporal object $T(t)$ and the total transmitted power B is recorded by a bucket detector, which is a slow, non-temporally resolving photodiode as in the following equation:

$$B = \int I(t)T(t)\,\mathrm{d}t. \tag{2}$$

The measurement is repeated N times, and the encryption is completed. In the transmission process, Alice only needs to transmit the parameter of the chaotic laser to Bob through the private channel and transmit the ciphertext to Bob through the common channel. This method improves the transmission efficiency than the previous methods. During the decryption, Bob uses a parameter of chaotic to generate chaotic laser and uses compressive sensing algorithm (CS) to decrypt the signal. The process is described in Eq. 3.

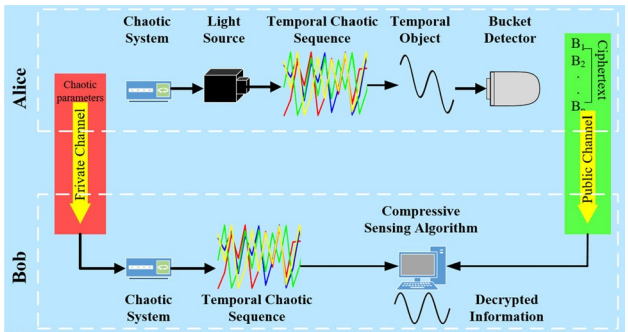


Fig. 1 Scheme of the method based on compressive temporal ghost imaging with chaotic laser

The CS theory states that if the image or signal is sparse, then under certain conditions on the observation operator, it can be recovered accurately from the compressive measurements, where $\|\bullet\|_1$ represents the L1-norm:

$$T_{cs} = \arg\min : \|T\|_1 \quad s.t. \ B = \int I(t)T(t)\,\mathrm{d}t. \tag{3}$$

2.2 Public key cryptography principle

Conventional optical encryption schemes are symmetrical, that is to say, encrypting and decrypting with the same key, optical encryption based on ghost imaging is also similar. It creates some problems. Both the key and the ciphertext are in the encipherer, and there is a key distribution problem. When the encipherer transmits the key to the receiver, the cost of the system will be increased whether the secret channel is established or additional security measures are taken. During the transmission, once the information is stolen, the security of plaintext is threatened. The emergence of public key cryptography provides us with new ideas for solving these problems.

Public key cryptography is an asymmetric encryption, that is, encryption and decryption use different keys. The transmitter uses a public key to encrypt plaintext, and the receiver decrypts the ciphertext with a private key. Public key cryptography mainly has the following principles: a public key only corresponds to a private key; the private key is known only to its owner, and the public key is available to anyone; if one of the keys is used for encryption, only the corresponding key can be used for decryption [28]. The process of public key cryptography is as follows:

1) Bob uses the public key cryptography algorithm to generate a pair of keys ($k_{\mathrm{pub}}, k_{\mathrm{pr}}$). Bob keeps the private key k_{pr} and sends the public key k_{pub} to Alice.
2) Alice uses the public key k_{pub} to encrypt the plaintext, get the ciphertext, and then transmit it to Bob.
3) Bob uses his private key to decrypt the ciphertext and obtains the plaintext (Fig. 2).

In this method, the private key is not transmitted, there is no risk of eavesdropping. The receiver generates the key, the key distribution problem is solved. At the same time, the ciphertext cannot be cracked even if the public key is stolen.

RSA public key algorithm is a good public key cryptosystem. Its secret principle is that it is impossible to perform prime factorization on a large composite number. This also shows that as long as the key is long enough, the RSA algorithm is basically impossible to crack, which is enough to ensure the security of the information [29]. The RSA key generation steps can be divided into the following six steps:

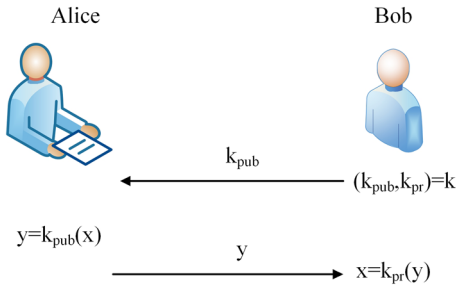


Fig. 2 The flowchart of the proposed public key cryptography

1) Select two large prime numbers p and q randomly
2) Calculate $N = p \times q$ and $L = \mathrm{lcm}\,(p-1, q-1)$.
3) Select an integer E, such that $1 < L$, $\gcd(E, L) = 1$
4) The decryption key D is calculated by $1 < D < L, E \times D \bmod L = 1$
5) (E, N) denotes the public key, and (D, N) denotes the private key.
6) Destroy p and q.

where 'lcm' represents lowest common multiple, ' gcd' represents the greatest common divisor, ' mod ' represents modulo operators.

The plaintext to be encrypted must be less than N, which is a prerequisite for correct decryption. We assume that the plaintext is m, the public key is (E, N), the ciphertext c is the remainder when m to $E-th$ power divided by N, the encryption operation is expressed as follows:

$$c = m^E \bmod N. \tag{4}$$

According to the same algorithm, the decryption operation can be expressed as follows:

$$m = c^D \bmod N. \tag{5}$$

3 Hybrid encryption scheme based on temporal ghost imaging

The traditional ghost imaging encryption scheme is a symmetric encryption method, but if the random patterns as the key are stolen, the security of the information is threatened. The public key cryptography is an asymmetric encryption method, which is more secure, but not suitable for encrypting long messages. In our study, we combine the ghost imaging and the public key cryptography to form a hybrid encryption scheme. The ghost imaging with chaotic laser encrypts the plaintext information, and the public key cryptography encrypts the parameters of chaotic laser. This method combines the advantages of symmetric encryption method and asymmetric encryption method to ensure

encryption efficiency and greatly improve the security of information. Compared to the traditional optical encryption method, this method solves the key distribution problem and reduces the additional cost of establishing the private channel while ensuring information security.

3.1 Encryption process

1) Bob uses the RSA public key algorithm to generate the private key (D, N) and the public key (E, N), then transmits the public key to Alice through the public channel.
2) Alice uses the pseudo-random number generator to generate the chaotic parameter P and generates the chaotic laser $I(t)$. The chaotic laser through the temporal object $T(t)$ and the total transmitted power is recorded by a bucket detector, which is a slow, non-temporally resolving photodiode, and the process is as Eq. 2. After multiple samplings, multiple values are obtained. The sequence of multiple numbers is ciphertext C_1.
3) Alice uses the public key (E, N) to encrypt the chaotic parameter P. Equation 4 shows the encryption method and the encrypted information C_2 is obtained.
4) Alice combines ciphertexts C_1 and C_2 to form a complete ciphertext. The encryption process is completed.

3.2 Decryption process

1) Bob receives Alice's ciphertexts and breaks it down into C_1 and C_2.
2) Bob uses the private key (D, N) to decrypt C_2 and obtain the chaotic parameter P; Eq. 5 shows the decryption method.
3) According to the chaotic parameter P, Bob controls the light source to generate the chaotic laser sequence $I(t)$ and records it.
4) The ciphertext C_1 and the chaotic sequence $I(t)$ stored sequentially are calculated using the compressive sensing algorithm (Eq. 3), and the result obtained is the plaintext $T'(t)$ after decryption (Fig. 3).

This optical hybrid encryption scheme combines advantages of the temporal ghost imaging encryption method and the public key cryptography. The RSA public key algorithm is simple and easy to operate, ensures the security of information without establishing a security channel, and solves the key distribution problem. Chaotic laser is used as a key to encrypt plaintext information, and chaotic laser is very sensitive to initial values. Therefore, the security of this system is guaranteed as long as the RSA algorithm ensures the security of the initial values of chaotic laser. At the same time, the problem of low efficiency of the RSA algorithm encrypting long information is solved.

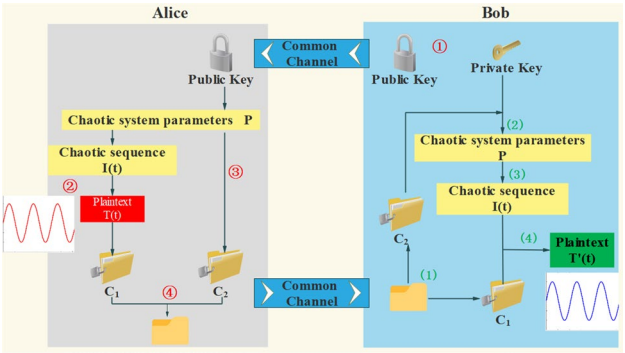


Fig. 3 Scheme of the proposed method

4 Numerical simulation

We use Matlab 2018a as the experiment platform to verify the feasibility and safety of this scheme. The chaotic model is a Logistic model. L1-regularized least square is utilized to recover the information. The sampling ratio is 100%, where the sampling ratio is defined as the ratio of the sampling numbers and the number of plaintexts.

4.1 Feasibility analysis

Here, we verify the feasibility of the encryption scheme; we encrypt a variety of signals, the signal interval is set to 0.02 s. To facilitate the calculation, select a simple prime number in the RSA algorithm. Specific steps are as follows:

1) Bob selects two prime numbers $p = 47$, $q = 61$ to obtain the public key $E = (167, 2867)$ and the private key $D = 1223$. Then, Bob transmits the public key to Alice.
2) Alice uses chaotic parameters which are generated by a pseudo-random number generator to obtain the chaotic laser. Alice uses temporal ghost imaging to encrypt the plaintext and uses the public key to encrypt chaotic parameters. The plaintexts are shown in the first row of Fig. 4. The ciphertexts are shown in the second row of Fig. 4.
3) Bob decrypts the received ciphertext using the private key and compressive sensing algorithm. The decrypted information is shown in the third row of Fig. 4.

Figure 4 shows that the plaintexts and ciphertexts are very different. In addition, the decrypted information and

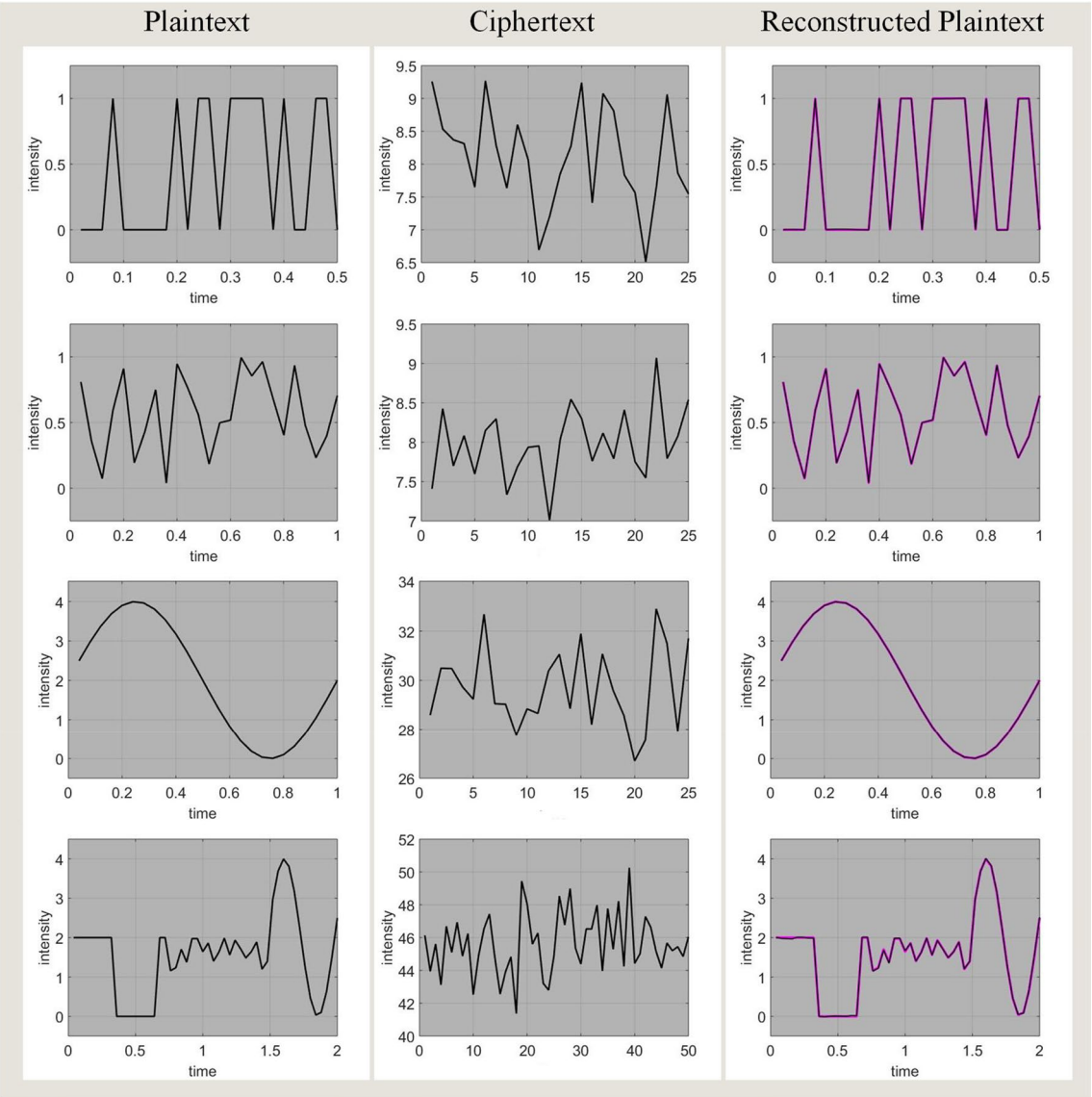


Fig. 4 Simulation results

the original information is the same. Simulation results verify the feasibility of this method.

4.2 Safety analysis

4.2.1 Ciphertext-only attack

The ciphertext-only attack is that the attacker analyzes the key or the plaintext corresponding to the ciphertext from the intercepted ciphertext. If the encryption method cannot resist a ciphertext-only attack, then this method is not safe. In this paper, the ciphertext is transmitted through the common channel and is easy to be stolen, so the correlation distribution of two adjacent pixels and histogram are used to verify the system's performance against ciphertext-only attacks [30].

First, the high-quality RSA algorithm ensures the security of chaotic parameters. The histogram is a method of analyzing the quality of the encryption algorithm. Different signals have different histogram distribution; it is a great method when the value of the ciphertext in the histogram is identical. We use the proposed method to encrypt different signals. The result is shown in Fig. 5. The result suggests that the original signals have different histogram distribution, while the encrypted signals are similar to each other, that is to say, the assailant is hardly obtaining useful information from the histogram.

Figure 6 shows that during the correlation of two adjacent pixels, signals show a regular distribution, it reflects the high

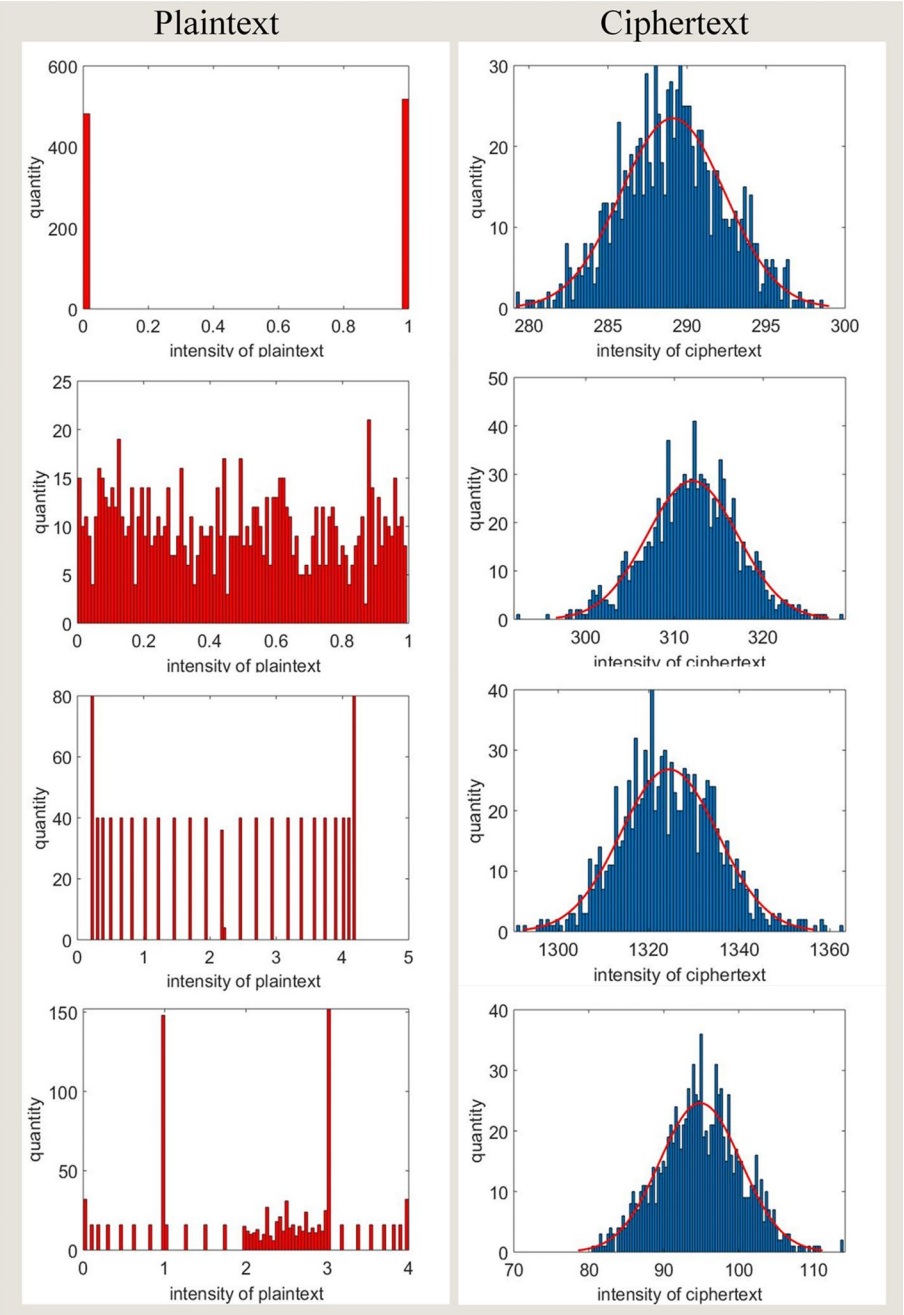


Fig. 5 Histogram of the different signals and the encrypted signals

correlation of two adjacent pixels. However, the ciphertext shows a disordered distribution; it reflects the weak correlation between two adjacent pixels. It shows that the assailant cannot use statistical analysis to obtain the plaintext. It means that this method has a strong resistance to ciphertext-only attack.

4.2.2 Chosen/known-plaintext attack

Chosen plaintext attack and known-plaintext attack mainly uses plaintext-ciphertext pairs to deduce the key until the system crack. Therefore, a secure encryption system must be able to withstand chosen/known-plaintext attack. To verify the security of HTGI, we use the chosen-plaintext attack and known-plaintext in paper [31, 32].

As shown in Fig. 7, for the traditional temporal ghost imaging encryption scheme (TGI), chosen/known-plaintext attacks can crack the encrypted information to a certain extent, but HTGI cannot be cracked. Since TGI as a symmetric encryption scheme is easy to be cracked, but HTGI as an asymmetric encryption scheme can effectively resist these attack types, especially the development of public key

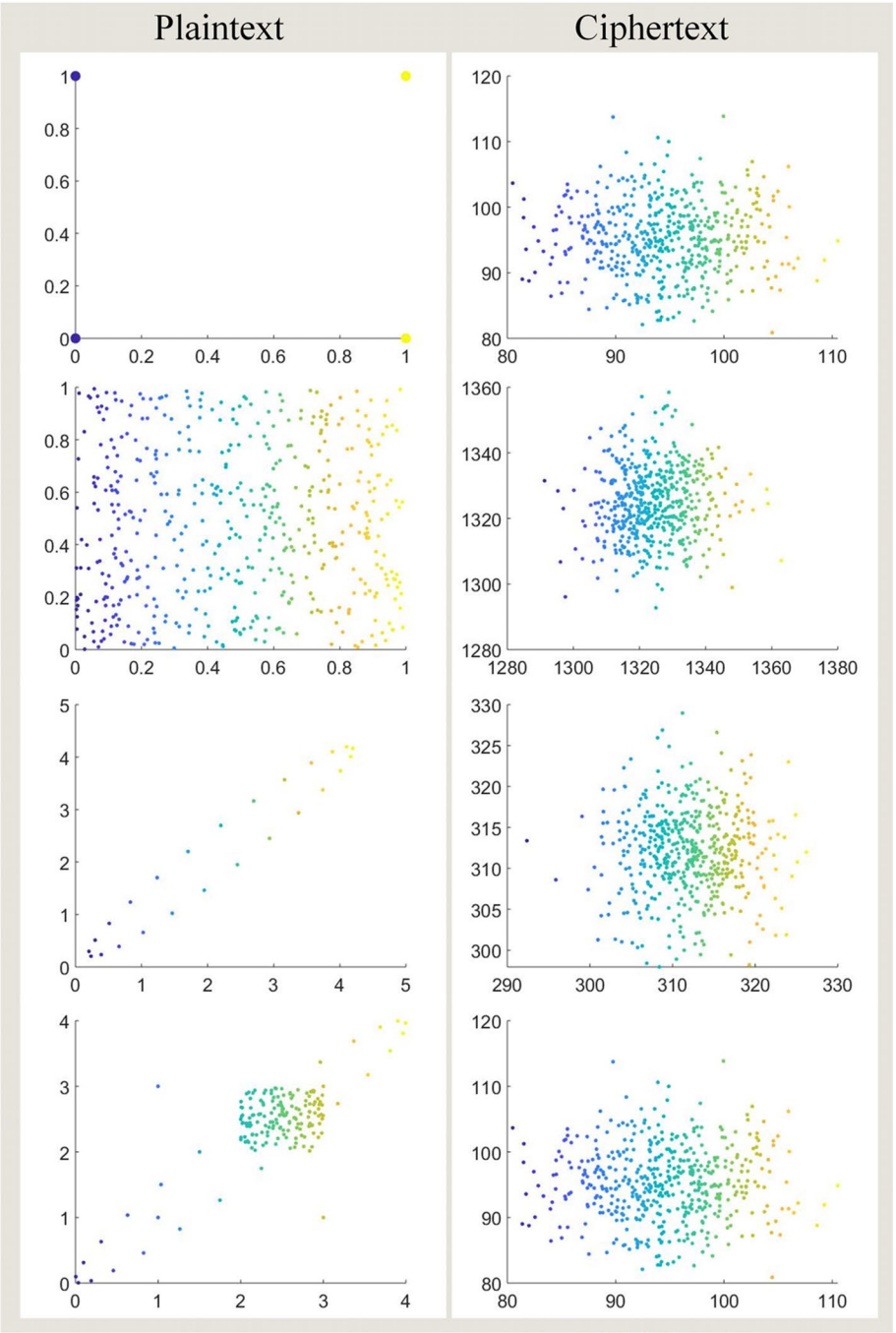


Fig. 6 Correlation distribution of two adjacent pixels

cryptography will also improve the system security. Notably, to facilitate display, the decrypted information is obtained after the threshold judgment operation. The premise of these two attack schemes is that the modulation patterns used as the key are fixed.

In password-cracking mode, differential attacks are also a type of chosen-plaintext attack. Therefore, a good encryption algorithm must have anti-differential attack capability, even if there is a slight difference between the plaintext, the acquired ciphertext is very different. The number of signals changing rate (NSCR), and unified average changing intensity (UACI) are usually used to measure the anti-differential attack ability. Assume that the amplitude of the two ciphertext signals at the i-th point is $C(i)$,$C'(i)$. If $C(i)$=$C'(i)$, and then $D(i) = 0$. If $C(i) \neq C'(i)$, then $D(i) = 1$. NSCR and UACI are defined as, where N is the number of plaintext signals:

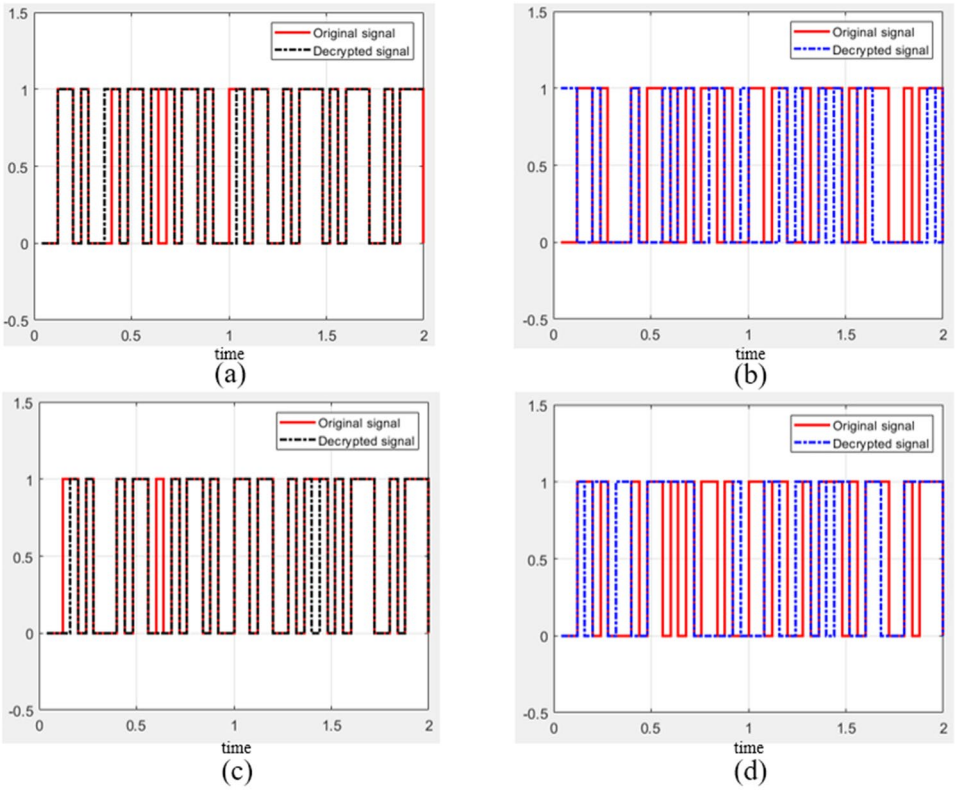


Fig. 7 Chosen/known-plaintext attack results. **a** Chosen-plaintext attack results of TGI; **b** chosen-plaintext attack results of HTGI; **c** known-plaintext attack results of TGI; **d** known-plaintext attack results of HTGI

$$\mathrm{NSCR} = \frac{\sum_{i=0}^{N-1} D(i)}{N} \times 100\%, \tag{6}$$

$$\mathrm{UACI} = \frac{1}{N} \sum_{i=1}^{N} \left| |C'(i)| - |C(i)| \right| \times 100\%. \tag{7}$$

The experiment selected 100 groups for encryption, and each group had 2 kinds of data, 1 being the original plaintext, the other 1 being the data that changes in a random position. By calculation, it can be found that their average values are $\overline{\mathrm{NSCR}} = 100\%$ and $\mathrm{UACI} = 6.44\%$. NSCR remains at 1 and UACI is lower than the standard. It shows that when the plaintext change slightly, the value of the whole ciphertext changes considerably, which indicates that the algorithm has a strong anti- differential attack ability.

Simulation results show that our method has a strong resistance to chosen-plaintext attack performance.

4.3 Robustness analysis

Ciphertext can suffer loss and noise during transmission, so we need to verify the robustness of this method.

4.3.1 Noise analysis

The ciphertext is disturbed by noise during its transmission, and it can affect the accuracy of decrypted information. We use the correlation coefficient (CC) as an evaluation standard. The closer to 1 is the value of the CC, the better is the similarity. When the value of CC is 1, it indicates that the two signals are identical. CC is defined as:

$$\mathrm{CC} = \frac{E\{[O - E(O)][D - E(D)]\}}{\sqrt{E\{[O - E(O)]^2\}}\sqrt{E\{[D - E(D)]^2\}}}, \tag{8}$$

where O and D stands for the plaintext and the recovered plaintext, respectively, $E\{\bullet\}$ stands for the expectation operator [22].

The Gaussian white noise is added to the ciphertext as shown below:

$$C' = C + x \cdot D, \tag{9}$$

where C' and C represent the noise ciphertext and the ciphertext, respectively. x is the coefficient indicating the noise strength, and D represents the Gaussian random data with zero-mean, and the variance is 1 [28].

The encryption system based on RSA and double random-phase encoding (RDRPE) in paper [28] is compared

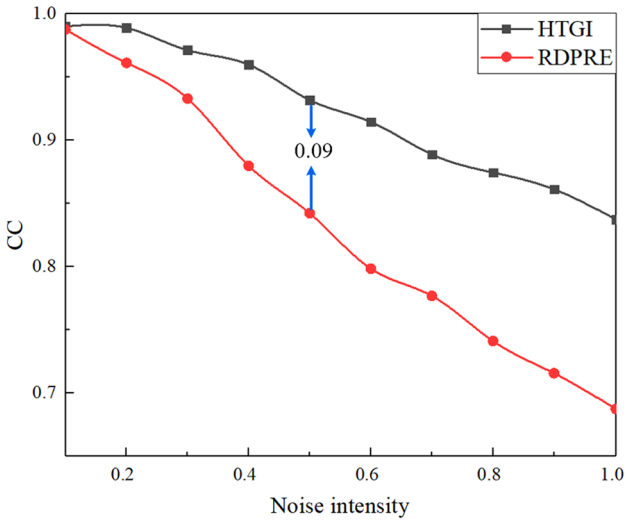


Fig. 8 Noise intensity and correlation coefficient

with HTGI. From the results of Fig. 8, the curve of CC trends down with the increase of noise intensity. However, HTGI is better than RDRPE in the same intensity. When the noise intensity is 0.5, the performance of our scheme is improved by 10% compared with RDRPE. Moreover, with the increase of noise intensity, the superiority of this scheme will be more and more obvious. When the noise intensity is 1, the decrypted information of HTGI still has a strong correlation with the original information.

4.3.2 Occlusion analysis

Loss of information can affect the accuracy of the decrypted information. The ciphertext is partly cut off to verify the robustness of this method. Figure 9 shows the corresponding images.

In Fig. 9, the accuracy of decrypted information is gradually reduced with the increase of occlusion ratio. When the occlusion ratio is 40%, the characteristics of plaintext in the reconstructed information can be seen. However, when the occlusion ratio is higher than 50%, some of the useful information from the reconstructed information cannot be obtained.

Random patterns are often used as modulation patterns for ghost imaging and temporal ghost imaging. Figure 10 shows the CC between the reconstructed information and the original information of random temporal ghost imaging (RTGI) and HTGI at different occlusion ratios. With the increase of occlusion ratio, the decrypted information quality of both decreased. However, HTGI is slightly stronger than RTGI at the same occlusion ratio. Overall, HTGI is robust to the information loss, and HTGI has a 3% improvement effect than RTGI.

4.3.3 Light intensity analysis

In the practice system, changing of light source can cause that Alice's chaotic sequence is different from Bob's, which affects the decryption result. Without considering other

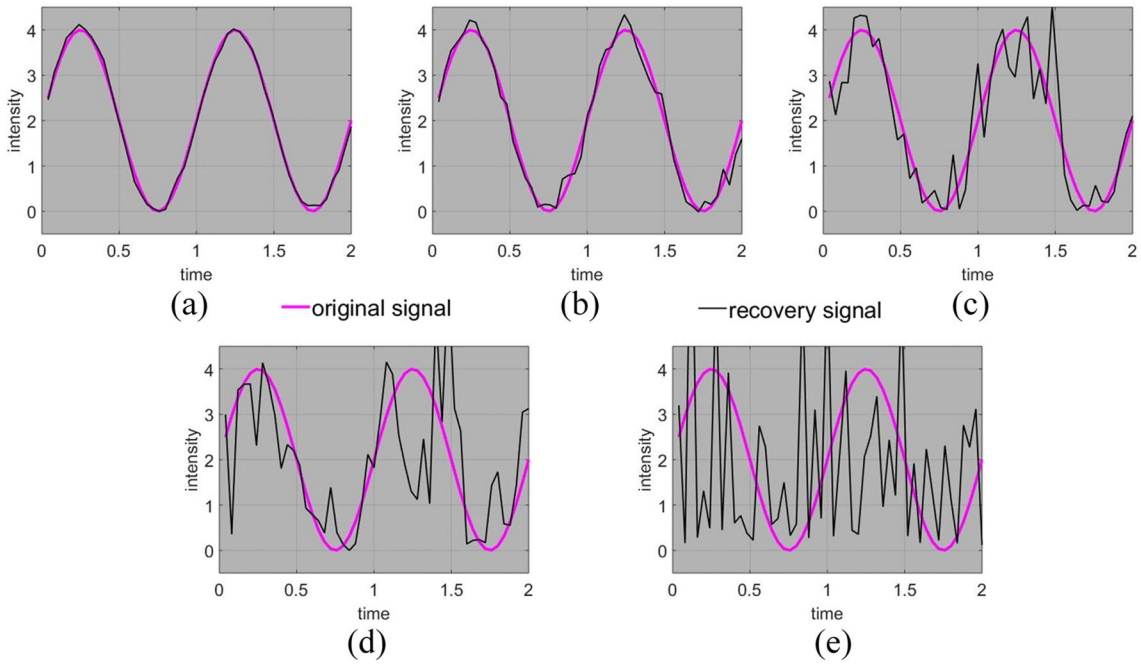


Fig. 9 Decrypted information at different occlusion ratios. **a**–**e** The occlusion ratios are 10%, 20%, 30%, 40%, and 50%, respectively

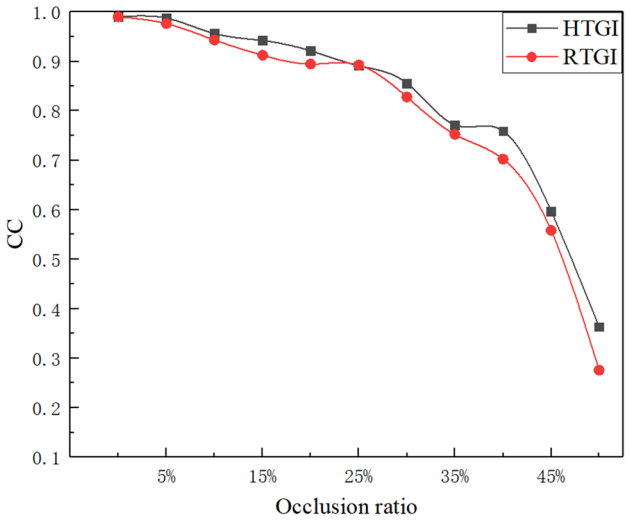


Fig. 10 Occlusion ratio and correlation coefficient

factors, the change of light source intensity will affect the intensity of chaotic sequence rather than the waveform. Next, we will explore the influence of light intensity on decryption by changing the intensity of chaotic sequence.

Figure 11 shows that the reconstructed signal strength increases with the decrease of intensity of chaotic sequence, and decreases with the increase of the intensity of chaotic sequence. Therefore, the reconstructed signal strength is negatively correlated with the intensity of chaotic sequence. But the intensity change of chaotic sequence does not affect the signal waveform. Therefore, even if the light intensity changes, the receiver can still get useful information.

4.4 Data size analysis

In this scheme, the private key is in the safe keeping of the receiver, so only the ciphertext needs to be transmitted. However, the traditional temporal ghost imaging encryption scheme needs to transmit a large number of modulation patterns (i.e., keys).

As shown in Table 1, for a given data, the amount of information to be transmitted by HTGI is much smaller than traditional temporal ghost imaging encryption schemes. More specifically, the data size of HTGI is approximately

Table 1 Data size

	Plaintext	Temporal ghost imaging	HTGI
Data size (byte)	1999	19,053,955	18,017

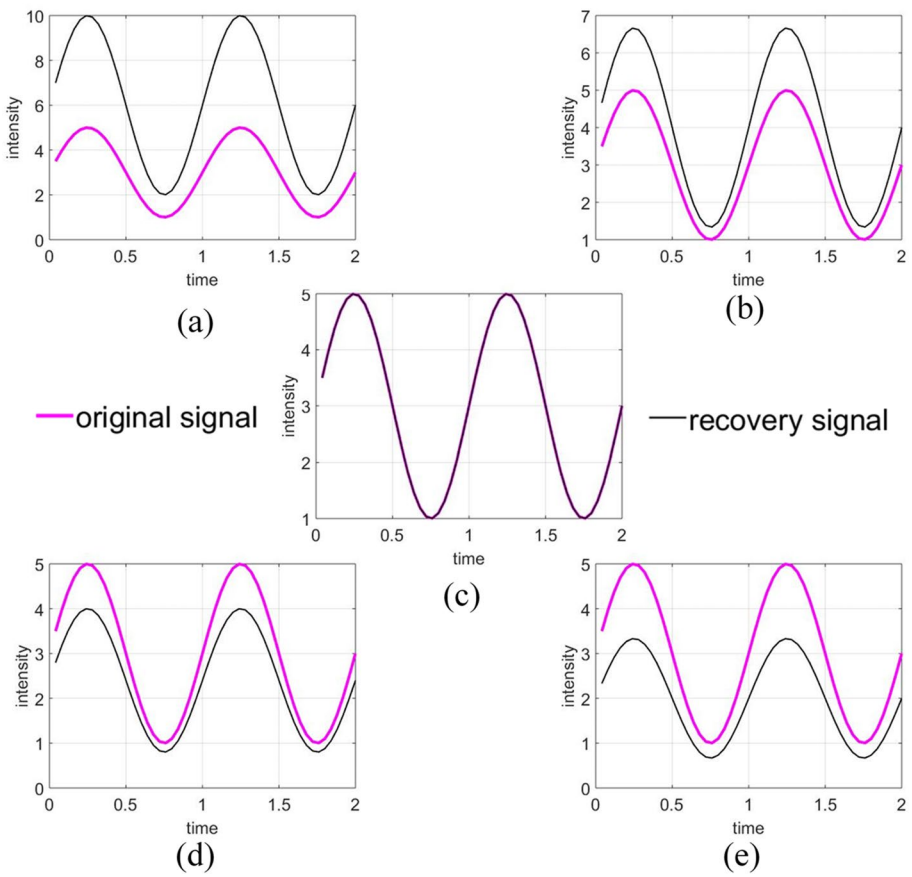


Fig. 11 Decrypted information in chaotic sequence of different intensities. **a**–**e** are reconstructed results with chaotic sequence strength of 50%, 75%, 100%, 125% and 150% of the original chaotic sequence strength

three orders of magnitude less than traditional temporal ghost imaging encryption scheme. Obviously, HTGI greatly reduces the amount of data transmitted, improves the transmission efficiency and the decryption speed at the same time.

5 Discussion and conclusion

In this study, we transform the ghost imaging encryption system from symmetric encryption system to hybrid encryption system, which greatly improves the security of the system. The effectiveness of temporal ghost imaging in optical fiber and wireless transmission [33, 34], and the advantage of being able to detect signals beyond the detector bandwidth [27], provide a new idea for realizing high-speed secure communication. However, it is worth noting that this scheme is a totally classical technology without connection to quantum key distribution, has limitations when coping with quantum computing or quantum eavesdroppers. This technique combined with quantum key distribution technology will provide a new idea to solve this problem. Similar to spatial ghost imaging, we need to run multiple times to achieve effective encryption and decryption of information, so exploring better temporal modulation patterns and reconstruction algorithms can also greatly improve the efficiency of the system.

In conclusion, we propose an optical hybrid encryption scheme based on the compressive temporal ghost imaging and the public key cryptography. This method combines asymmetric public key cryptography and symmetric temporal ghost imaging encryption to ensure security while improving transmission efficiency. In this scheme, the key is generated by the receiver, so the problem of key distribution is solved. The encipherer uses temporal ghost imaging with chaotic laser to encrypt plaintext information and uses RSA public key algorithm to encrypt chaotic laser initial parameters, both of them form ciphertext. The encipherer only transmitted ciphertexts, reducing the extra cost of establishing a private channel. The receiver uses the private key to obtain the chaotic laser initial parameters and then uses the compressive sensing algorithm to obtain the plaintext information. Simulations verify the effectiveness of this scheme and show strong robustness and anti-attack ability. Compared with traditional temporal ghost imaging encryption methods, our method greatly reduces the data size and improves efficiency. In a word, optical hybrid encryption based on compressive temporal ghost imaging and public key cryptography can achieve high-quality, high-safety, wide-applicability and low-cost encryption.

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